Enhanced Object Tracking Based on an Improved CSRT Framework with Kalman Filtering and Deep Feature Refinement

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Abstract

Visual object tracking remains a challenging problem in computer vision due to target appearance variation, partial occlusion, scale change, fast motion, and background clutter. Although the Channel and Spatial Reliability Tracker (CSRT) provides a robust and efficient correlation-filter baseline by exploiting spatial reliability masks and channel-wise discriminative filters, its response-driven localization and fixed update strategy may still lead to drift when the target undergoes abrupt motion, ambiguous appearance changes, or unreliable response peaks.

In this paper, we propose a Confidence-Aware CSRT framework, termed CA-CSRT, which enhances the standard CSRT tracking pipeline with Kalman filtering and lightweight deep feature refinement. The proposed framework first uses a Kalman filter to predict a motion-guided search region for the target. CSRT-based correlation localization is then performed within this constrained region, and the quality of the resulting response map is evaluated using peak-based confidence information. This confidence score is used to adaptively fuse the CSRT measurement with the Kalman prediction and to regulate model updates, reducing the risk of model contamination under low-confidence observations. In addition, a lightweight deep refinement component is incorporated to improve scale and appearance stability without relying on full-frame detection at every frame.

Experiments on OTB100 and an evaluated LaSOT subset are conducted to analyze the tracking accuracy, precision, runtime efficiency, and FPS–AUC trade-off of the proposed framework. The results indicate that CA-CSRT improves the robustness and average success performance over the pure CSRT baseline in the evaluated benchmarks while maintaining practical real-time or near-real-time runtime depending on the deployment platform, making it suitable for lightweight visual tracking and embedded deployment scenarios.

Keywords: Visual Object Tracking, CSRT, Correlation Filter, Kalman Filtering, Confidence-Aware Tracking, Deep Feature Refinement, Embedded Vision.

1 Introduction

Visual object tracking is a fundamental problem in computer vision, aiming to estimate the position and scale of a target object throughout a video sequence given only its initial state. Despite continuous progress in recent years, robust tracking in real-world scenarios remains challenging due to fast motion, partial occlusion, scale variation, illumination change, background clutter, and target appearance deformation. These challenges become more critical in real-time and embedded applications, where tracking algorithms must maintain a practical balance between robustness, accuracy, and computational efficiency.

Correlation filter-based trackers have been widely adopted in online visual tracking because they provide an attractive trade-off between tracking accuracy and runtime performance. Among them, the Channel and Spatial Reliability Tracker (CSRT) improves discriminative correlation filtering by introducing spatial reliability masks and channel-wise weighting, allowing the tracker to suppress background interference more effectively than earlier correlation filter methods. However, the standard CSRT framework remains mainly response-driven. When the response map becomes ambiguous due to abrupt motion, occlusion, or distractors, the tracker may rely on unreliable local maxima and may update its model using contaminated observations. In addition, fixed update strategies and handcrafted feature representations can limit the adaptability of CSRT under long-term tracking and scale variation.

To address these limitations, this paper proposes a Confidence-Aware CSRT framework, termed CA-CSRT. The proposed framework enhances the CSRT-family tracking pipeline by integrating Kalman-based motion prediction, response-confidence estimation, confidence-guided measurement fusion, and lightweight deep feature refinement. Instead of relying only on the current CSRT response, CA-CSRT first predicts a motion-guided search region using a Kalman filter. The CSRT-based localization result is then evaluated using peak-quality information from the response map. This confidence score is used to adaptively fuse the visual measurement with the motion prediction and to regulate model updates, reducing the risk of tracking drift when the response map is unreliable.

The proposed framework is designed with both robustness and runtime efficiency in mind. By restricting visual localization to a motion-guided region and avoiding full-frame detection at every frame, CA-CSRT aims to improve tracking stability while preserving real-time or near-real-time performance. In addition, lightweight deep feature refinement is incorporated to improve scale and appearance stability. This design makes the framework suitable for practical tracking scenarios, including embedded deployment on resource-constrained platforms such as Jetson Nano.

The main contributions of this work are summarized as follows:

- We propose CA-CSRT, a confidence-aware CSRT framework that combines correlation-filter localization with Kalman-based motion prediction.
- We introduce a response-confidence mechanism to adaptively fuse CSRT measurements with motion prediction and to control model updates under unreliable observations.
- We incorporate lightweight deep feature refinement to improve scale and appearance stability without relying on full-frame detection at every frame.
- We evaluate the proposed framework on OTB100 and an evaluated LaSOT subset using success AUC, precision, FPS, and FPS–AUC trade-off analysis, and discuss its practical deployment on Jetson Nano.

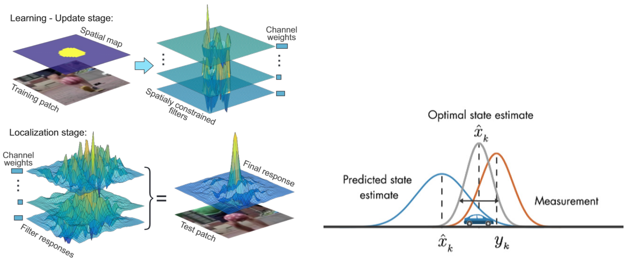
2 Related Works

2.1 Correlation Filters and Motion Modeling

Correlation filter-based trackers have been widely studied in online visual object tracking because they provide a favorable balance between localization accuracy and computational efficiency. Early methods such as MOSSE [1] and KCF [2] formulate tracking as a discriminative filtering problem and exploit efficient Fourier-domain operations for fast response computation. Subsequent methods extend this formulation with multi-channel features, scale estimation, spatial regularization, and adaptive model update strategies.

The Channel and Spatial Reliability Tracker (CSRT) [3] is a representative correlation-filter tracker that improves target localization by estimating spatial reliability masks and channel-wise reliability weights. These mechanisms help suppress background interference and emphasize discriminative feature channels. However, CSRT remains mainly response-driven: the target state is inferred from the peak of the correlation response map, and the appearance model is updated according to a predefined strategy. When the response map becomes flat, noisy, or multi-modal due to fast motion, occlusion, or background distractors, relying only on the response peak can lead to inaccurate localization and model drift.

Motion modeling provides a complementary prior for visual tracking. Classical filters, including Kalman filters and particle filters, estimate target states over time by combining a motion prediction with noisy observations. The Kalman filter [5] is particularly suitable for lightweight tracking systems because it provides an efficient recursive prediction-correction mechanism under a linear Gaussian assumption. In CA-CSRT, the Kalman prediction is used to guide the search region and to provide a motion prior when the CSRT response is unreliable. Instead of using a fixed fusion rule, the proposed framework evaluates the response quality and uses this confidence information to decide how strongly the tracker should trust the visual measurement or the motion prediction.



(a) CSRT overview (b) Kalman motion model

Figure 1: Background components related to CA-CSRT: (a) the Channel and Spatial Reliability Tracker and (b) Kalman filter-based motion prediction.

2.2 Deep Feature-Based Tracking

Deep feature representations have been increasingly adopted in visual tracking because they can capture more semantic and discriminative information than handcrafted features. Modern deep trackers often achieve strong accuracy by using Siamese networks, transformer-based architectures, or heavy feature backbones. However, these approaches may introduce significant computational overhead and can be difficult to deploy on resource-constrained platforms.

Hybrid tracking approaches aim to balance robustness and efficiency by integrating deep representations into classical tracking pipelines rather than replacing the entire tracker with a heavy end-to-end model. For correlation-filter trackers, deep features can be used as auxiliary information for appearance representation, scale estimation, or bounding-box refinement. Scale variation is particularly important because inaccurate target size estimation directly affects success scores based on intersection-over-union. Classical scale-adaptive methods such as DSST [4] address this issue through dedicated scale estimation, while lightweight refinement modules can further improve scale and appearance stability within local search regions.

The proposed CA-CSRT framework follows this hybrid design principle. It uses CSRT-family correlation localization as an efficient baseline, Kalman prediction as a lightweight motion prior, response confidence to regulate measurement fusion and model update, and lightweight deep refinement to improve scale and appearance stability. This combination is intended to improve the robustness-efficiency trade-off while remaining practical for real-time and embedded visual tracking.

3 Proposed Method

3.1 Overview of the Proposed CA-CSRT Framework

Given an initial target bounding box $B_0 = (x_0, y_0, w_0, h_0)$, the goal of visual object tracking is to estimate the target state $B_k = (x_k, y_k, w_k, h_k)$ for each incoming frame k. In this work, we formulate tracking as a confidence-aware closed-loop estimation problem, where correlation-based localization, motion prediction, response reliability estimation, scale refinement, and model update are jointly considered.

The proposed CA-CSRT framework follows a motion-guided and confidence-controlled pipeline. First, a Kalman filter predicts the target state from previous frames and provides a motion prior for the current frame. This prediction is used to define a constrained search region around the expected target location, reducing the search ambiguity caused by abrupt motion or background distractors. Within this region, CSRT-based correlation localization is performed to obtain a visual measurement from the response map.

Since the reliability of the CSRT response may vary significantly across frames, CA-CSRT explicitly evaluates the quality of the response map using peak-based confidence information, particularly the Average Peak-to-Correlation Energy (APCE). A sharp and unambiguous response peak indicates a reliable visual measurement, whereas a flat or multi-modal response suggests uncertainty caused by occlusion, distractors, or appearance change. The resulting confidence score is then used to adaptively fuse the CSRT measurement with the Kalman prediction.

After obtaining the fused target center, a lightweight deep refinement component is applied within the motion-guided region to refine the target scale and aspect ratio. The final bounding box is then fed back to update the tracker. Unlike a fixed update strategy, CA-CSRT regulates the model update according to the estimated confidence level: reliable observations are used to update the appearance model, whereas low-confidence observations lead to conservative updates to reduce model contamination and tracking drift.

Overall, CA-CSRT enhances the standard CSRT-family tracking pipeline by combining motion prediction, response-confidence evaluation, adaptive measurement fusion, scale refinement, and confidence-controlled model update in a unified framework. The overall architecture of the proposed CA-CSRT pipeline is illustrated in Figure 2.

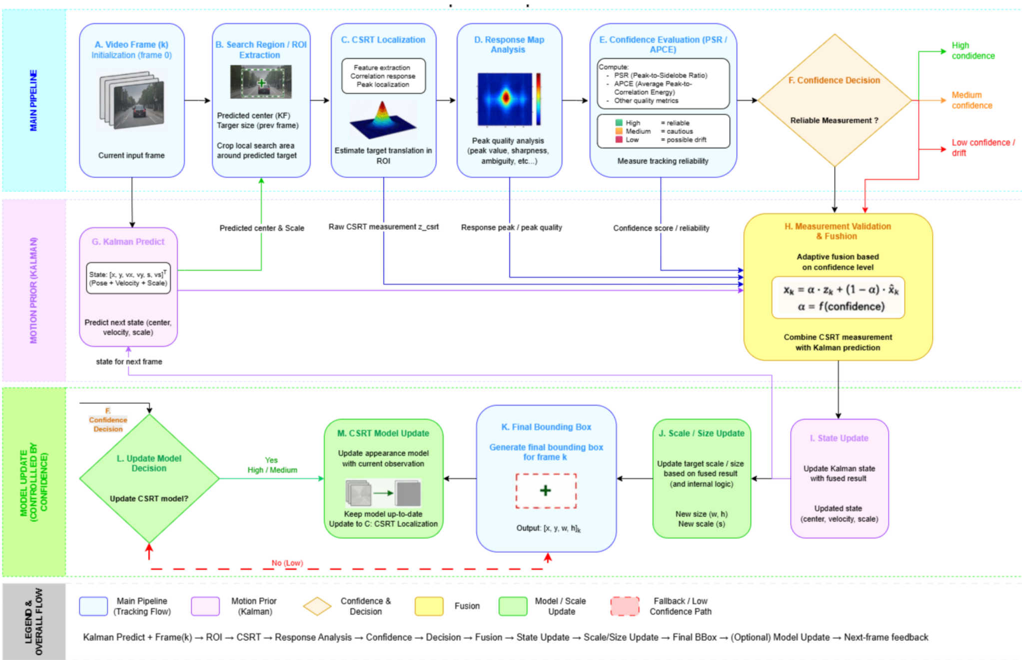


Figure 2: Overall architecture of the proposed CA-CSRT pipeline, including motion prediction, CSRT localization, response-confidence evaluation, adaptive fusion, state/scale update, and confidence-controlled model update.

3.2 Correlation-Based Localization Using CSRT

Within the motion-guided search region predicted in the previous step, CA-CSRT applies a CSRT-family correlation localization module to obtain a visual measurement of the target center. Rather than treating this measurement as the final tracking result, CA-CSRT uses it as an observation whose reliability must be evaluated before fusion with the motion prior.

Let $x_k \in \mathbb{R}^{H \times W \times C}$ denote the multi-channel feature representation extracted from the search region at frame k. The CSRT localization process can be abstracted as learning channel-wise correlation filters $\{f_c\}_{c=1}^{C}$ with a spatial reliability constraint:

$$\min_{\{f_c\}} \sum_{c=1}^{C} \|m \odot (x_{k,c} * f_c - y)\|^2 + \lambda \|f_c\|^2, \tag{1}$$

where m denotes the spatial reliability mask, y is the desired Gaussian-shaped response, λ is the regularization parameter, and $*$ denotes correlation.

Given the learned filters, the response map for the current search region is computed as a weighted combination of channel responses:

$$R_k = \sum_{c=1}^{C} w_c \cdot (x_{k,c} * f_c), \tag{2}$$

where w_c is the reliability weight of the c-th feature channel. The peak location of the response map provides the raw CSRT center estimate:

$$c_k^{\mathrm{CSRT}} = \arg\max_p R_k(p), \tag{3}$$

which is represented as the measurement vector $z_k^{\mathrm{CSRT}} = [x_k^{\mathrm{CSRT}}, y_k^{\mathrm{CSRT}}]^\top$.

Although the response peak provides an efficient localization cue, its reliability can vary significantly under fast motion, occlusion, deformation, or background distraction. A sharp single peak usually indicates a confident localization, whereas a flat or multi-modal response may indicate ambiguity. Therefore, CA-CSRT does not directly update the target state from z_k^{CSRT}. Instead, the response map R_k is further analyzed to estimate measurement reliability, which is then used for confidence-guided Kalman fusion and model update.

3.3 APCE-Based Measurement Reliability Estimation

The CSRT response map R_k provides not only the peak location but also useful information about the reliability of the localization result. In CA-CSRT, this reliability is estimated using the Average Peak-to-Correlation Energy (APCE), which measures the sharpness and contrast of the response map. Given the maximum and minimum response values $R_k^{\max}$ and $R_k^{\min}$, APCE is defined as:

$$\mathrm{APCE}_k = \frac{\left(R_k^{\max} - R_k^{\min}\right)^2}{\frac{1}{|R_k|}\sum_i \left(R_k(i) - R_k^{\min}\right)^2 + \epsilon}, \tag{4}$$

where $|R_k|$ denotes the number of response locations and ϵ is a small constant used for numerical stability.

A high APCE value indicates that the response map contains a sharp and dominant peak, suggesting that the CSRT measurement is likely reliable. In contrast, a low APCE value often corresponds to a flat, noisy, or multi-modal response map, which may occur under occlusion, fast motion, appearance deformation, or background distractors. Therefore, APCE is used as an internal confidence cue rather than as an external benchmark metric.

To use this reliability estimate in the tracking pipeline, the raw APCE score is converted into a normalized confidence coefficient:

$$\alpha_k = \sigma\left(\beta(\mathrm{APCE}_k - \tau)\right), \tag{5}$$

where $\sigma(\cdot)$ is the sigmoid function, τ is the confidence threshold, and β controls the transition sharpness. The coefficient $\alpha_k \in [0, 1]$ represents the confidence of the CSRT measurement at frame k. This coefficient is subsequently used to control the fusion between the CSRT measurement and the Kalman prediction, and to support conservative model update when the response confidence is low.

3.4 Kalman Filter with APCE-Guided Measurement Fusion

3.4.1 State Representation and Motion Prediction

CA-CSRT uses a lightweight constant-velocity Kalman filter to provide a motion prior for the target center. The target state is defined as:

$$\mathbf{s}_k = [x_k, y_k, \dot{x}_k, \dot{y}_k]^\top, \tag{6}$$

where (x_k, y_k) denotes the target center and $(\dot{x}_k, \dot{y}_k)$ denotes its velocity. The prediction step is written as:

$$\hat{\mathbf{s}}_{k|k-1} = \mathbf{F}\mathbf{s}_{k-1}, \tag{7}$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}\mathbf{P}_{k-1}\mathbf{F}^\top + \mathbf{Q}, \tag{8}$$

where $\mathbf{F}$ is the state transition matrix, $\mathbf{P}_{k|k-1}$ is the predicted state covariance, and $\mathbf{Q}$ is the process noise covariance.

The predicted center $\hat{c}_{k|k-1} = (\hat{x}_{k|k-1}, \hat{y}_{k|k-1})$ is used to generate the motion-guided search region for CSRT localization. This motion prior reduces the search ambiguity when the target undergoes fast displacement or when the local background contains distractors.

3.4.2 Confidence-Guided Measurement Update

The raw CSRT center estimate obtained from the response map is treated as a visual measurement:

$$\mathbf{z}_k^{\mathrm{CSRT}} = [x_k^{\mathrm{CSRT}}, y_k^{\mathrm{CSRT}}]^\top. \tag{9}$$

Instead of using a fixed measurement noise covariance, CA-CSRT adapts the covariance according to the APCE-based confidence coefficient α_k:

$$\mathbf{\Sigma}_k^z = \frac{1}{\alpha_k + \epsilon}\mathbf{\Sigma}_0^z, \tag{10}$$

where $\mathbf{\Sigma}_0^z$ is the nominal measurement noise covariance and ϵ is a small constant for numerical stability. When α_k is high, the covariance becomes smaller and the CSRT measurement is trusted more. When α_k is low, the covariance increases, causing the update to rely more conservatively on the motion prediction.

The Kalman gain is computed as:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}^\top(\mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^\top + \mathbf{\Sigma}_k^z)^{-1}, \quad (11)$$

where $\mathbf{H}$ maps the state vector to the measured target center. The fused state is then updated by:

$$\hat{\mathbf{s}}_k = \hat{\mathbf{s}}_{k|k-1} + \mathbf{K}_k(\mathbf{z}_k^{\mathrm{CSRT}} - \mathbf{H}\hat{\mathbf{s}}_{k|k-1}). \quad (12)$$

This APCE-guided update allows CA-CSRT to adaptively balance visual localization and motion prediction. Reliable response maps lead to stronger correction from the CSRT measurement, while ambiguous responses result in more conservative updates to reduce drift.

3.5 Deep Feature–Based Scale Refinement

After the target center is obtained from APCE-guided Kalman fusion, the remaining uncertainty mainly lies in the target scale and aspect ratio. In CA-CSRT, a lightweight deep refinement component is used as a local auxiliary module to refine the bounding-box size within the motion-guided region. Unlike full-frame detection, this module operates only on a constrained region around the estimated target location, which helps limit computational overhead.

Let Ω_k denote the local region cropped around the fused target center at frame k. The refinement module estimates the relative scale variation with respect to the previous target size. Instead of directly regressing the absolute width and height, the regression target is represented using log-scale ratios:

$$y_w = \log \frac{w_k}{w_{k-1}}, \qquad y_h = \log \frac{h_k}{h_{k-1}}. \quad (13)$$

The predicted scale ratios are denoted as (μ_w, μ_h), and the refined target size is recovered as:

$$w_k^{\mathrm{ref}} = w_{k-1}\exp(\mu_w), \qquad h_k^{\mathrm{ref}} = h_{k-1}\exp(\mu_h). \quad (14)$$

To avoid abrupt scale changes, CA-CSRT applies a temporal smoothing step before producing the final bounding-box size:

$$w_k = (1-\eta_k)w_{k-1}+\eta_k w_k^{\mathrm{ref}}, \qquad h_k = (1-\eta_k)h_{k-1}+\eta_k h_k^{\mathrm{ref}}, \quad (15)$$

where $\eta_k \in [0,1]$ controls the scale update rate. In practice, η_k can be fixed or adjusted according to the reliability of the refinement output. This local refinement strategy complements the CSRT response-based localization by improving scale stability while preserving the lightweight nature of the overall tracking pipeline.

3.6 Closed-Loop Model Update

The final bounding box:

$$B_k = (x_k, y_k, w_k, h_k) \quad (16)$$

is used to update the CSRT model.

The update rate is implicitly regulated by APCE, preventing model contamination when response confidence is low.

Through this closed-loop design, **APCE serves as the central link between correlation-based localization and motion estimation**, enabling robust and stable tracking under challenging conditions.

4 Experimental Results

4.1 Experimental Setup

All experiments are conducted following the standard **One-Pass Evaluation (OPE)** protocol used in visual object tracking.

Datasets

We evaluate the proposed method on the **OTB-100** benchmark, which contains a wide range of challenging scenarios including fast motion, scale variation, occlusion, background clutter, and illumination change.

Compared Trackers

To analyze the contribution of each component, we conduct both ablation studies and comparative evaluations with representative trackers provided by the OTB toolkit.

All trackers use the same initialization and are evaluated under identical conditions.

Evaluation Metrics

Tracking performance is measured using:

- **Precision**: center location error
- **Success**: Intersection-over-Union (IoU)–based success rate (AUC)
- **Success**: Frames Per Second (FPS)

4.2 Result Evaluation on OTB-100

4.2.1 Success Evaluation (AUC)

The success metric evaluates the overlap between the predicted bounding box B_k and the ground-truth bounding box B_k^{gt} using Intersection-over-Union (IoU):

$$\mathrm{IoU}_k = \frac{B_k \cap B_k^{\mathrm{gt}}}{B_k \cup B_k^{\mathrm{gt}}} \quad (17)$$

For a given threshold $t \in [0,1]$, the success rate is defined as the percentage of frames satisfying:

$$\mathrm{IoU}_k > t \quad (18)$$

The **Area Under the Curve (AUC)** is computed by integrating the success rate over all thresholds:

$$\mathrm{AUC} = \int_0^1 \mathrm{Success}(\mathrm{IoU} > t)\, dt \quad (19)$$

Results and analysis.

AUC is the primary evaluation metric in this work, as it jointly reflects localization accuracy and scale stability. The proposed method achieves the highest AUC among all compared trackers, demonstrating its robustness under scale variation and appearance change.

The improvement in AUC mainly originates from the deep scale refinement module and the uncertainty-guided EMA, which effectively stabilize bounding box size and suppress abrupt scale fluctuations.

4.2.2 Precision Evaluation

Precision evaluates the localization accuracy of a tracker by measuring the center location error between the predicted bounding box and the ground-truth bounding box.

Specifically, precision is defined as the percentage of frames in which the Euclidean distance between the predicted target center c_k and the ground-truth center c_k^{gt} is within a predefined threshold τ:

$$\text{Precision}(\tau) = \frac{1}{N}\sum_{k=1}^{N} \mathbf{1}(\|c_k - c_k^{gt}\| < \tau) \tag{20}$$

where N denotes the total number of frames and τ is set to 20 pixels following the OTB evaluation protocol.

Results and analysis.

The baseline CSRT tracker achieves competitive precision under moderate motion but suffers from degraded localization accuracy in the presence of fast motion and background clutter.

By incorporating Kalman Filter-based motion prediction, CSRT+KF improves center localization stability by exploiting temporal motion continuity.

Furthermore, the APCE-guided Kalman update effectively suppresses unreliable correlation responses, leading to additional precision gains.

The full model achieves the highest precision among all compared trackers, demonstrating that confidence-aware motion estimation plays a critical role in accurate target localization.

4.3 Discussion

Overall, the experimental results demonstrate that the proposed APCE-guided Kalman filtering framework significantly improves tracking robustness.

By integrating deep scale refinement and uncertainty-aware temporal smoothing, the proposed tracker achieves superior performance while preserving real-time capability.

5 Conclusions

In this paper, we proposed a robust visual tracking framework that enhances the classical CSRT tracker by explicitly integrating motion modeling, response reliability estimation, and deep scale refinement within a unified pipeline.

The proposed method introduces an **APCE-guided Kalman filtering strategy**, which adaptively fuses correlation-based measurements with motion prediction. By quantifying the reliability of the CSRT response map, unreliable updates are effectively suppressed, leading to improved robustness under fast motion, background clutter, and ambiguous appearances.

To address scale variation, a lightweight **deep scale refinement module** is incorporated. The module is trained offline following the GOT-10k sampling protocol and predicts relative scale changes in a local and motion-stabilized region. Furthermore, an **uncertainty-guided exponential moving average (EMA)** is employed during inference to stabilize scale updates and reduce temporal jitter.

Extensive experiments on the OTB benchmark demonstrate that the proposed tracker achieves consistent improvements over the baseline CSRT tracker, particularly in terms of **success score (AUC)**, which is the primary evaluation metric in this work. Ablation studies further confirm that each component—Kalman filtering, APCE-based reliability estimation, deep scale refinement, and EMA smoothing—contributes positively to the overall tracking performance.

Overall, the proposed framework improves tracking accuracy and stability while preserving real-time capability. Future work will explore tighter integration between motion estimation and appearance modeling, as well as extending the framework to more challenging long-term tracking scenarios.

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